

What is Ansible?

If you've ever had to configure dozens of servers by hand — logging into each one, running the same commands, hoping nothing goes wrong — you already understand the problem Ansible was built to solve. Ansible is an open-source **IT automation tool** developed by Red Hat that lets you automate repetitive tasks across your infrastructure from a single control machine.

What makes Ansible stand out is its simplicity. It uses **YAML-based playbooks** — plain-English-style scripts — to define what you want your systems to look like. There's no agent software to install on target machines; Ansible communicates over **SSH (or WinRM for Windows)**, making it lightweight and easy to adopt. It is **idempotent** by design, meaning you can run the same playbook multiple times without fear of breaking things — it only makes changes when needed.

Main Use Cases

■ Configuration Management

Keep servers in a consistent, desired state. Whether managing 5 or 5,000 machines, Ansible ensures every host is configured the same way — right packages installed, right configs in place, every time.

■ Application Deployment

Roll out new versions of applications across multiple environments (dev, staging, prod) in a repeatable and controlled way — far less error-prone than manual deployments.

■ Cloud Provisioning

Spin up and configure cloud resources on AWS, Azure, GCP, and others. Ansible's cloud modules make Infrastructure as Code (IaC) straightforward without requiring a dedicated tool like Terraform for simpler setups.

■ Orchestration

Coordinate complex multi-tier workflows — for example, drain a load balancer, update the app servers, run database migrations, then bring traffic back — all in the right order, automatically.

■ Security & Compliance

Automate security hardening tasks: applying patches, enforcing password policies, disabling unnecessary services, and auditing configurations to stay compliant with standards like CIS or NIST.

■ CI/CD Pipeline Integration

Ansible integrates naturally with Jenkins, GitLab CI, GitHub Actions, and others, acting as the deployment engine in modern DevOps pipelines.

Why Ansible? — Quick Snapshot

Agentless	Human-Readable	Large Ecosystem	Free & Open Source
No software to install on managed nodes — just SSH.	YAML playbooks read almost like plain English.	3,000+ modules for cloud, network, OS, apps, and more.	Core Ansible is free; Red Hat offers enterprise support.